

1. Low volume turns AVP on

WHAT YOU SEE

Banner: LOW EFFECTIVE VOLUME TURNS AVP ON

WHAT IT ENCODES

Hypovolemia is a NON-osmotic (baroreceptor-mediated) stimulus for ADH/AVP: when effective circulating volume drops, carotid/aortic baroreceptors override the normal osmotic shutoff and the posterior pituitary keeps releasing ADH even though serum osmolality is already low. ADH then drives free-water retention at the V2 receptor (aquaporin-2 insertion in the collecting duct), diluting serum Na. This is the unifying mechanism for ALL hypovolemic hyponatremia: true Na/water loss → low volume → ADH stays on → water retained > Na. Mnemonic: 'volume beats osmolality' — the body defends pressure before it defends tonicity.

BOARD TRAP

Q stem shows a dry, tachycardic, orthostatic patient with Na 124 and the answer choice says 'ADH should be suppressed.' WRONG — in hypovolemia ADH is appropriately ON despite low osm because the baroreceptor (volume) stimulus dominates the osmotic one. Discriminator: dry mucous membranes, flat neck veins, BUN/Cr >20, high uric acid and high BUN all confirm volume depletion driving the ADH.

2. Split by urine Na

WHAT YOU SEE

Bottom banner: HYPOVOLEMIC SPLIT BY U Na THEN GIVE NS

WHAT IT ENCODES

Once true hypotonic hyponatremia + hypovolemia is established, the single best next test is URINE SODIUM: <20 mEq/L = extrarenal loss (kidney is appropriately Na-avid), >20 mEq/L = renal loss (kidney is inappropriately wasting Na). Treatment is the same category — isotonic 0.9% NaCl (154 mEq/L Na) to restore volume — but cause-finding differs. Once volume is repleted the baroreceptor stimulus disappears, ADH switches off, and a brisk water diuresis begins, so anticipate and cap overcorrection at <8 mEq/L/24h.

BOARD TRAP

Stem gives a hypovolemic patient and asks for the BEST initial diagnostic step; distractors include 'fluid restrict' or 'check TSH/cortisol first.' WRONG — fluid restriction worsens hypovolemia, and the highest-yield discriminating test is urine Na (then urine Cl). Order osmolality → volume → urine Na; restriction is for euvolemic/hypervolemic states, not the dry patient.

3. Extrarenal gate UNa<20

WHAT YOU SEE

Left locked door: U Na 20 EXTRARENAL

WHAT IT ENCODES

Extrarenal losses (GI, skin, third-space) → the kidney's response is appropriate: it maximally conserves Na, so urine Na is <20 mEq/L (often <10) and FENa <1%. Causes: vomiting, diarrhea, sweating, burns, pancreatitis/ileus third-spacing. The 'locked door' = nephron clamping shut on Na. Urine is also concentrated (high Uosm) because ADH is on.

BOARD TRAP

Stem: patient with profuse diarrhea, Na 126, urine Na 8 — a choice labels this 'renal salt wasting' or 'SIADH.' WRONG — a LOW urine Na means the kidney is working correctly and conserving Na, i.e., the loss is extrarenal. SIADH has urine Na typically >30 and is euvolemic, not dry.

4. Renal gate UNa>20

WHAT YOU SEE

Right door: U Na RENAL (>20)

WHAT IT ENCODES

Renal losses → urine Na inappropriately >20 mEq/L despite hypovolemia (the kidney is the leak). Causes: thiazide/loop diuretics, cerebral salt wasting, primary adrenal insufficiency (Addison), salt-losing nephropathy, type 2 (proximal) RTA bicarbonaturia, ketonuria, and osmotic diuresis (glucose, mannitol, urea). The 'unlocked door' = nephron can't hold Na. Stop the offending drug, replace Na/volume with NS, and screen cortisol/aldosterone if cause unclear.

BOARD TRAP

Hypovolemic patient with urine Na 45 and a distractor says 'extrarenal GI loss.' WRONG — high urine Na in a dry patient means the kidney itself is the source (renal salt wasting). Top board causes to scan for: thiazides (most common drug), Addison's (look for hyperkalemia + hypotension), and CSW after recent CNS event.

5. Vomiting

WHAT YOU SEE

Figure labeled VOMIT at far left

WHAT IT ENCODES

Vomiting is an extrarenal cause but a classic exception on urine studies: loss of gastric HCl → metabolic alkalosis → filtered bicarbonate spills into urine and obligates Na excretion (to maintain electroneutrality), so urine Na can be PARADOXICALLY >20 early even though the patient is volume-depleted. The unmasking test is urine CHLORIDE, which is low (<20) and correctly reveals the volume-avid kidney and chloride-responsive alkalosis.

BOARD TRAP

Vomiting patient, Na 128, urine Na 35 → distractor calls it 'renal salt wasting.' WRONG — in vomiting the high urine Na is driven by bicarbonaturia, not true renal wasting. Discriminator: check urine Cl (<20 = volume depletion / chloride-responsive metabolic alkalosis). Urine Na alone is unreliable in the vomiting patient.

6. Diarrhea

WHAT YOU SEE

Figure labeled DIARRHEA

WHAT IT ENCODES

Diarrhea = extrarenal GI loss of isotonic-to-hypotonic fluid; the kidney conserves Na, so urine Na <20 and urine Cl is also low. Diarrhea typically produces a NON-anion-gap (hyperchloremic) metabolic acidosis from stool bicarbonate loss — the mirror image of vomiting's alkalosis — useful for telling the two GI causes apart.

BOARD TRAP

Stem pairs hyponatremia + non-gap metabolic acidosis + low urine Na and a choice says 'type 2 RTA.' Be careful — both diarrhea and proximal RTA give non-gap acidosis, but diarrhea has LOW urine Na (volume-avid) and an appropriately NEGATIVE urine anion gap, whereas RTA wastes Na renally with urine Na >20.

7. Sweat / burns

WHAT YOU SEE

Sun and figure labeled SWEAT, burning house labeled BURNS

WHAT IT ENCODES

Sweat and burns lose hypotonic fluid through the skin (sweat Na ~30–60 mEq/L, hypotonic relative to plasma) → free-water and Na loss with effective hypovolemia and appropriate renal Na avidity (urine Na <20). Extensive burns lose plasma into the wound/interstitium (a third-space effect too). Replace with isotonic crystalloid; burn resuscitation uses weight/%BSA-based formulas (e.g., Parkland) rather than treating the Na number in isolation.

BOARD TRAP

Marathon runner or burn patient with hyponatremia and a choice attributes it to 'SIADH.' WRONG — these are hypovolemic skin losses with a dry exam and low urine Na; the fix is volume/Na repletion, not fluid restriction. (Exercise-associated hyponatremia is actually most often from excessive hypotonic fluid intake — still NOT a restriction-then-no-volume answer.)

8. Pancreatitis / third space

WHAT YOU SEE

Pot labeled PANCREATITIS THIRD SPACE

WHAT IT ENCODES

Third-spacing (acute pancreatitis, bowel obstruction/ileus, peritonitis, major trauma, crush injury) sequesters large volumes of fluid into the gut wall, peritoneum, or interstitium. Intravascular (effective) volume falls even though total body water may be normal/high → baroreceptor-driven ADH → hyponatremia with avid Na retention (urine Na <20). Treat the underlying process and replete intravascular volume with isotonic fluid.

BOARD TRAP

Pancreatitis patient looks puffy/third-spaced and a choice says 'hypervolemic — fluid restrict.' WRONG — the EFFECTIVE arterial volume is LOW (fluid is sequestered, not circulating), so this behaves as hypovolemic hyponatremia needing volume, not restriction. Judge by effective circulating volume, not total body water.

9. Thiazide (renal)

WHAT YOU SEE

Pill figure labeled THIAZIDE

WHAT IT ENCODES

Thiazides (HCTZ, chlorthalidone) are the #1 DRUG cause of hyponatremia. They block the Na-Cl cotransporter (NCC) in the distal convoluted tubule — the diluting segment — so the kidney loses Na yet, crucially, can still CONCENTRATE urine via an intact medulla/ADH, impairing free-water excretion. Volume contraction also adds a baroreceptor ADH stimulus. Classic victim: small, elderly woman started on a thiazide; can develop severe hyponatremia within days to weeks. Loop diuretics cause it far less because they blunt the medullary gradient.

BOARD TRAP

Elderly woman recently started on HCTZ, Na 118, looks near-euvolemic; distractor is 'SIADH.' WRONG — thiazide hyponatremia masquerades as euvolemic but is a renal salt-loser with mild volume depletion. The discriminator/treatment is STOP the thiazide (and the picture, including low uric acid sometimes, can mimic SIADH) — you would never diagnose SIADH without first removing a thiazide.

10. Addison (adrenal)

WHAT YOU SEE

Figures labeled ADDISON

WHAT IT ENCODES

Primary adrenal insufficiency (Addison): mineralocorticoid (aldosterone) deficiency → renal Na wasting (urine Na >20), hyperkalemia, and a non-gap or mild metabolic acidosis; glucocorticoid (cortisol) deficiency → loss of cortisol's tonic inhibition of CRH/ADH → ADH excess → water retention. The combination produces hypovolemic hyponatremia + HYPERkalemia + hypotension, often with hyperpigmentation and an adrenal-crisis presentation. Treat with stress-dose hydrocortisone and IV NS.

BOARD TRAP

Hyponatremia + HYPERkalemia + hypotension and a choice says 'SIADH.' WRONG — SIADH is euvolemic with NORMAL or low K; hyperkalemia points away from SIADH and toward adrenal insufficiency. Confirm with a cosyntropin (ACTH stimulation) test and give hydrocortisone — don't fluid-restrict an Addisonian crisis.

11. CSW cerebral salt wasting

WHAT YOU SEE

Sign: CSW CEREBRAL SALT WASTE

WHAT IT ENCODES

Cerebral salt wasting: after a CNS insult (subarachnoid hemorrhage, TBI, neurosurgery, meningitis) the kidney inappropriately wastes Na (urine Na >20–40), causing HYPOVOLEMIC hyponatremia — the patient is genuinely DRY (negative fluid balance, low CVP, weight loss). Mechanism implicated: natriuretic peptide (BNP/'brain' natriuretic) excess ± sympathetic dysregulation. Treatment is OPPOSITE to SIADH: give salt and volume (isotonic or hypertonic saline, fludrocortisone), NOT restriction.

BOARD TRAP

Patient 5 days post-SAH with Na 125, urine Na 60; choices are SIADH (restrict fluids) vs CSW (give salt/volume). The board discriminator is VOLUME STATUS: CSW = hypovolemic/dry (give salt & fluid); SIADH = euvolemic (restrict). Choosing fluid restriction in a truly hypovolemic CSW patient worsens cerebral perfusion and is the classic wrong answer.

12. RTA / bicarb / ketonuria

WHAT YOU SEE

Signs: RTA BICARB, KETONURIA, OSMOTIC

WHAT IT ENCODES

Anion-driven (osmotic) renal Na loss: filtered anions that can't be reabsorbed obligate Na excretion to stay electroneutral — bicarbonate (proximal/type-2 RTA, or post-vomiting alkalosis), ketoanions (DKA, alcoholic/starvation ketosis), and osmotic diuresis (glucosuria, mannitol, high urea). Result: renal hypovolemic hyponatremia with urine Na >20. Fix the underlying anion problem and replete volume.

BOARD TRAP

DKA patient with hyponatremia and urine Na high — a choice calls it 'SIADH.' WRONG twice: (1) the ketoanion/osmotic diuresis is causing renal Na loss with volume depletion, and (2) hyperglycemia itself causes translocational/pseudo-low Na — you must CORRECT the measured Na for glucose (+~1.6–2.4 mEq/L per 100 mg/dL >100) before calling it true hyponatremia.

13. K high + Na low = adrenal

WHAT YOU SEE

Sign: K HIGH PLUS Na LOW EQUALS ADRENAL

WHAT IT ENCODES

Memory hook: in a hypovolemic patient, HYPERkalemia + hyponatremia = think ADRENAL (primary adrenal insufficiency / hypoaldosteronism). Aldosterone normally drives Na reabsorption and K/H secretion in the distal nephron; without it you waste Na (low Na) and retain K (high K) and H⁺ (acidosis). Next steps: AM cortisol, cosyntropin stim, renin/aldosterone; treat with hydrocortisone + NS.

BOARD TRAP

The hyperkalemia gets blamed on 'renal failure' or 'SIADH.' WRONG — with normal renal function, the Na-low/K-high pairing is the adrenal-insufficiency signature. SIADH keeps K normal/low (dilution), and pure volume depletion alone doesn't usually raise K unless aldosterone is deficient. Don't skip the ACTH-stimulation test.

14. Give NS, watch overcorrection

WHAT YOU SEE

IV bag 0.9 NS and clock: AVP OFF WATCH OVERCORRECTION

WHAT IT ENCODES

0.9% NaCl (154 mEq/L) restores effective volume and removes the baroreceptor ADH stimulus. The danger: once volume is repleted, ADH switches OFF and the kidney dumps the retained free water as a brisk dilute diuresis (autocorrection), so serum Na can rise FAST. Cap the rise at <8 mEq/L/24h (4–6 in high-risk: alcoholic, malnourished, hypokalemic, Na <105, liver disease). Monitor Na q2–4h and urine output; if rising too fast, give D5W ± DDAVP.

BOARD TRAP

After NS the urine output surges and Na climbs 10–12 mEq in a few hours; a choice says 'continue NS, this is expected recovery.' WRONG — that pace risks osmotic demyelination syndrome (ODS / central pontine myelinolysis). Switch to free water (D5W) and add DDAVP to re-lower/clamp the diuresis; the goal is a CONTROLLED <8 mEq/24h rise, not maximal speed.

15. Urate high in hypovolemia

WHAT YOU SEE

Coin/sun labeled URATE HIGH at top right

WHAT IT ENCODES

Serum URIC ACID is a slick discriminator: hypovolemia → avid proximal tubular reabsorption of urate (and BUN) → HIGH uric acid (often >4 mg/dL) and elevated BUN/Cr ratio. SIADH does the opposite — volume expansion suppresses proximal reabsorption → LOW uric acid (<4 mg/dL, FEurate >12%) and low/low-normal BUN. So high urate + high BUN points to a dry, ADH-driven hypovolemic state.

BOARD TRAP

Two near-identical euvolemic-appearing stems differ only by uric acid; a choice ignores it. Use it: LOW uric acid (<4) + low BUN supports SIADH/euvolemia; HIGH uric acid + high BUN supports hypovolemia (or post-correction in any cause). It's a high-yield, cheap test to separate SIADH from volume depletion when the exam is equivocal.